Predicting Weekly Work Hours

Micro-Project 4: Neural Networks

https://github.com/mascon0423/ANA500/tree/main/week4

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Problem Statement

- Week 3 predicted weekly work hours from age, education, occupation, and workclass. SVR improved on the mean baseline.
- The remaining problem: shorter workweeks were overpredicted and longer workweeks were underpredicted.
- Question: Can neural networks and additional demographic information improve overall predictions and errors at these extremes?
- Continue with the same Adult dataset, numerical work-hour outcome, test records, and grouped cross-validation framework.

Hypothesis Formulation

- A neural network using additional household and demographic information may improve predictions compared with Week 3's four-feature SVR.
- Possible explanation: nonlinear relationships and added context may capture differences in work hours that the earlier models missed.
- Full support requires lower overall test RMSE and lower MAE for both under-35 and over-40 hour workweeks. Report partial improvements separately.
- Include a recurrent neural network (RNN) to address the assignment. Compare it with a multilayer perceptron (MLP) and choose using validation results.

Acquire

- Use the unchanged Adult CSV from Weeks 1-3: 32,561 records and 15 columns from the UCI training sample.
- The source describes the 1994 Census database. Weekly work hours are numerical; this project predicts hours, not income categories.
- Keep source row order and verify the CSV checksum. Save split membership and reference results from Week 3.
- Tools: Jupyter Notebook, Python, Pandas, NumPy, Matplotlib, scikit-learn, and PyTorch.
- Becker & Kohavi (1996). Adult. https://doi.org/10.24432/C5XW20

Prepare: Original and Expanded Inputs

Feature set	Predictors	Purpose
Core: 4	Age, education, occupation, workclass	Direct continuation of Week 3
Expanded: 7	Core plus marital status, relationship, sex	Test whether added context improves predictions
Excluded	Hours or hour groups, income, fnlwgt, duplicate education code	Avoid outcome leakage; keep unweighted sample scope

Historical demographic associations do not establish causes or justify individual employment decisions.

Retain original numerical age and hours. Missing workclass and occupation become Unknown. Use each record equally.

Source: UCI Adult; unweighted historical sample. Same Week 3 partitions and grouped CV folds.

Prepare: Same Evaluation Framework

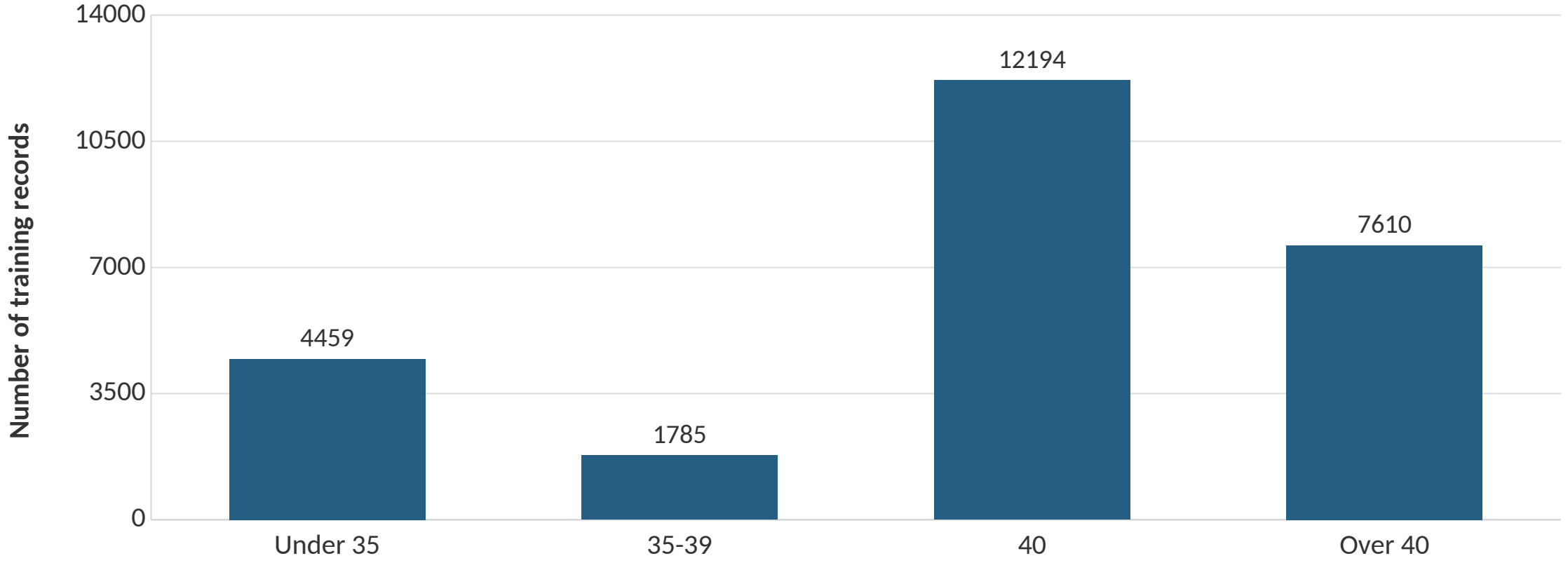
Stage	Records / folds	Use
Training	26,048 records	Fit transformations and models
Cross-validation	Same 3 grouped folds	Choose model and settings by mean RMSE
Inner stopping split	80/20 within each fold's training portion	Choose neural epochs without using the scored fold
Test	Same 6,513 records	Evaluate fixed choices and inspect errors

Exact repeated full profiles stay together in every split.

The earlier test results motivated this follow-up. This is an internal comparison on familiar records, not a new independent validation sample.

Source: UCI Adult; unweighted historical sample. Same Week 3 partitions and grouped CV folds.

Prepare: Work-Hour Distribution



The concentration at 40 hours makes it important to examine short and long workweeks separately. Groups here describe training records only.

Source: UCI Adult; unweighted historical sample. Same Week 3 partitions and grouped CV folds.

Analyze: Neural Network Architectures

Model	Selected hidden size	Final epochs	Parameters
MLP - core	32, then 16	33	1,793
RNN - core	2 recurrent layers of 16; dense 16	37	1,425
MLP - expanded	32, then 16	19	2,177
RNN - expanded	2 recurrent layers of 32; dense 16	37	4,449

Tested two sizes per neural family and feature set; no claim that these are optimal architectures.

Both families use the same information for each feature set. Every model predicts one continuous value: weekly work hours.

Source: UCI Adult; unweighted historical sample. Same Week 3 partitions and grouped CV folds.

Analyze: Why Include an RNN?

- I included an RNN to address the assignment instruction and compare it with a dense MLP.
- The Adult dataset has no natural time sequence. The RNN processes a fixed order of fields within one person's record.
- Each step contains encoded field values and a field identifier. The hidden state resets for each person; people are never linked into a sequence.
- The order is artificial. Retrain the preferred RNN with reversed and shuffled field orders to examine sensitivity using training cross-validation.
- Model choice should follow the data structure, prediction target, and validation evidence. An assigned architecture is not assumed to be best.

Analyze: Training and Model Selection

- Use the same three grouped CV folds for all methods. Fit scaling and category encoding only within the relevant training portion.
- Neural models: Adam, learning rate 0.001, weight decay 0.001, batches of 256, maximum 60 epochs, and patience 8.
- An inner grouped split chooses the epoch. Then refit the whole outer training portion for that fixed duration before scoring the outer fold.
- Refit each chosen neural configuration on all training rows using the median epoch from its three folds. Use seed 42 and deterministic CPU operations.
- The expanded SVR tests the original C/epsilon grid. The four Week 3 models reproduce their earlier results using the saved settings.

Analyze: Cross-Validation Results

Model	CV RMSE (hours)	Fold SD
Mean baseline	12.340	0.240
Linear - core	11.542	0.199
SVR - core	10.926	0.187
Tree - core	11.021	0.167
SVR - expanded	10.707	0.166
MLP - core	10.833	0.182
RNN - core	10.851	0.187
MLP - expanded	10.667	0.171
RNN - expanded	10.673	0.154

MLP - expanded is preferred by CV RMSE. This choice is fixed before test scoring. Fold SD describes three-fold variation, not a confidence interval.

Source: UCI Adult; unweighted historical sample. Same Week 3 partitions and grouped CV folds.

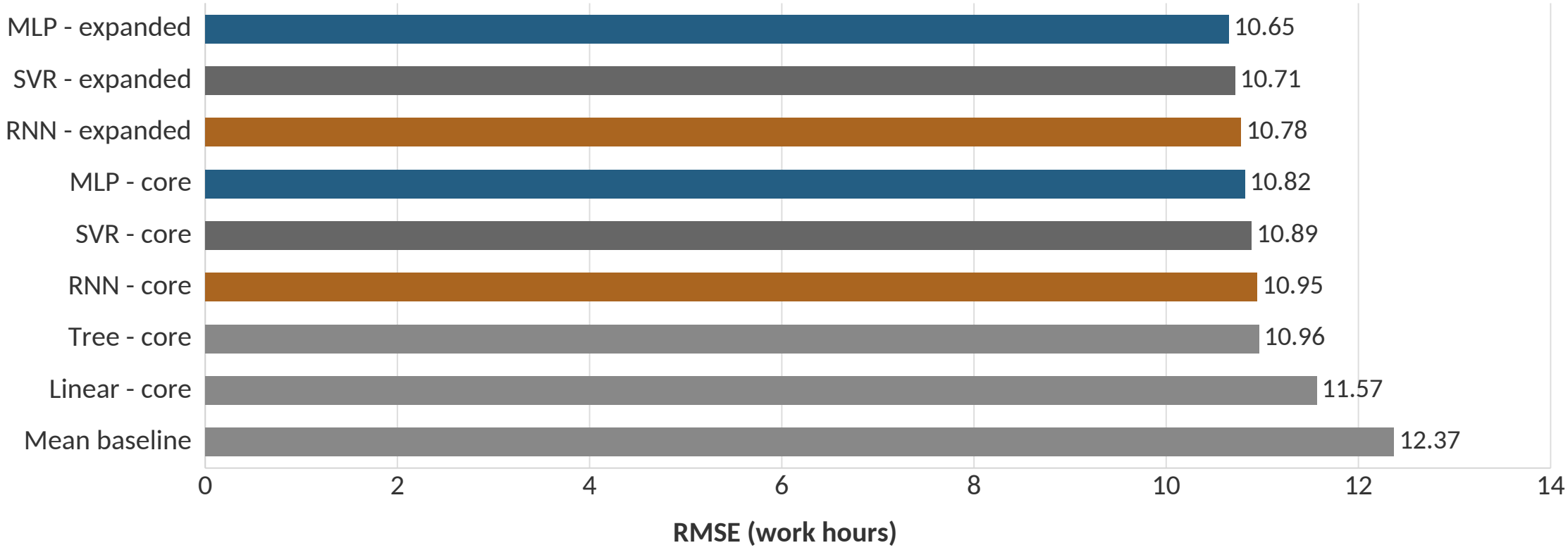
Report: Test Results

Model	MAE (hours)	RMSE (hours)	R-squared
Mean baseline	7.61	12.37	-0.000
Linear - core	7.87	11.57	0.126
SVR - core	7.16	10.89	0.225
Tree - core	7.49	10.96	0.214
SVR - expanded	7.07	10.71	0.250
MLP - core	7.34	10.82	0.235
RNN - core	7.36	10.95	0.217
MLP - expanded	7.25	10.65	0.259
RNN - expanded	7.30	10.78	0.240

MAE = mean absolute error; RMSE = root mean squared error. Both are in hours; lower is better. All models use the same 6,513 test records.

Source: UCI Adult; unweighted historical sample. Same Week 3 partitions and grouped CV folds.

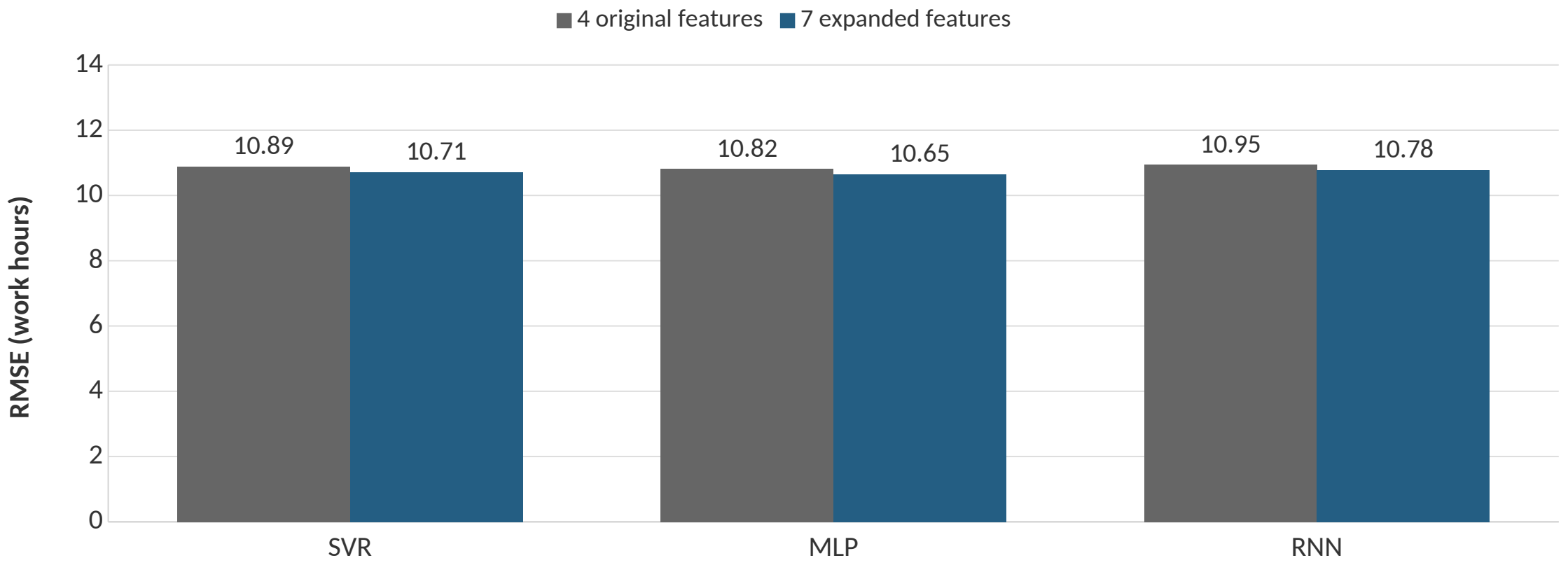
Report: Test RMSE (Work Hours)



The CV choice remains MLP - expanded. Test rankings describe performance on these records and do not trigger another round of tuning.

Source: UCI Adult; unweighted historical sample. Same Week 3 partitions and grouped CV folds.

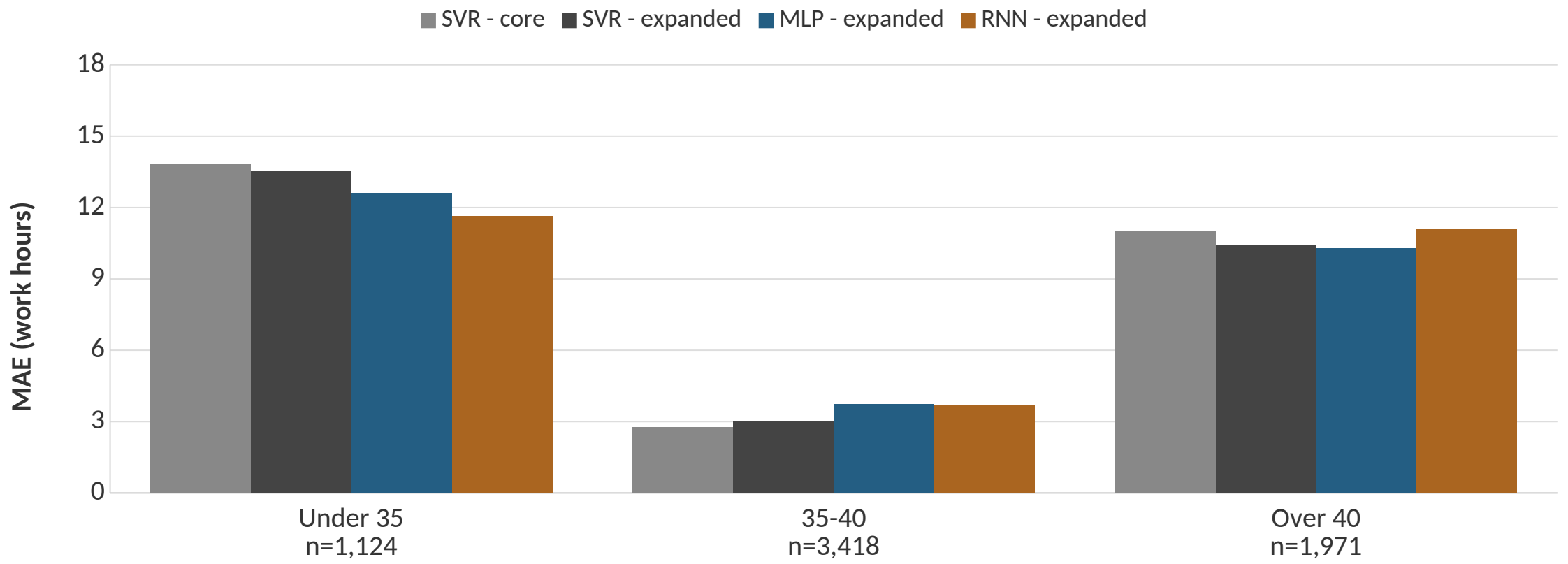
Report: Additional Features



Compare feature sets within each family. An improvement from expanded inputs alone does not establish that a neural architecture caused the gain.

Source: UCI Adult; unweighted historical sample. Same Week 3 partitions and grouped CV folds.

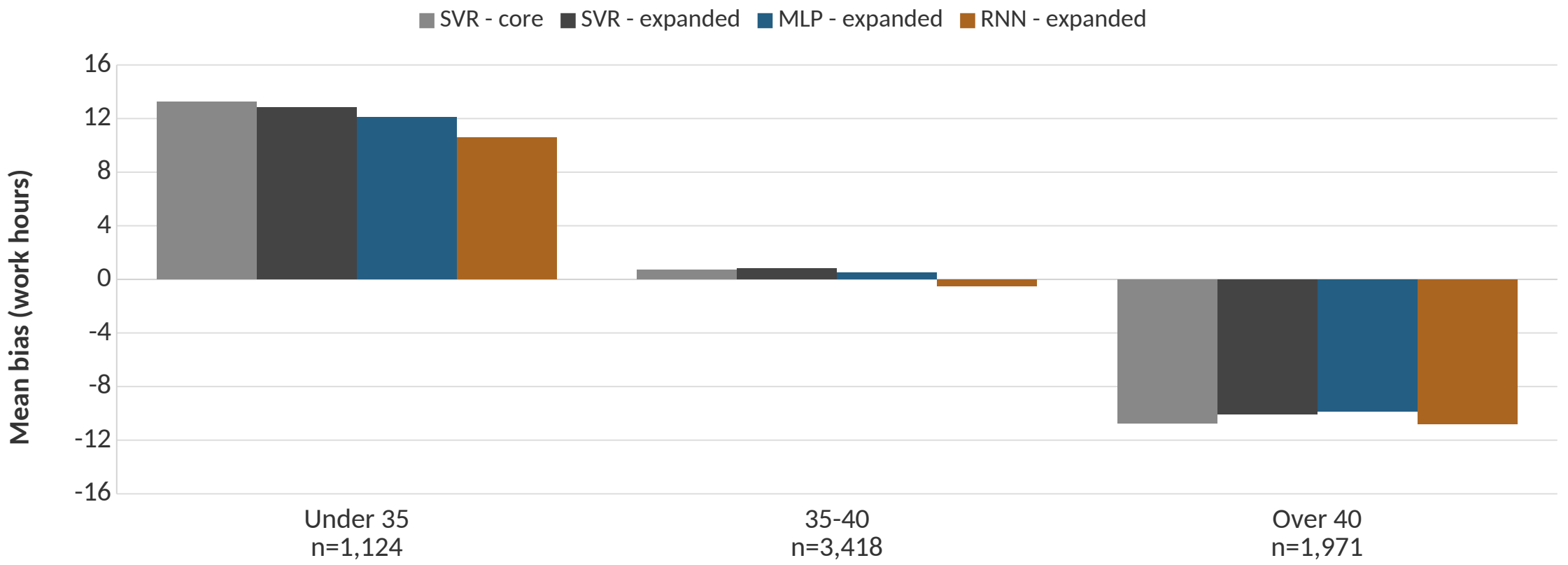
Report: Errors at the Extremes



Groups are defined by actual test hours for diagnosis only. Lower MAE means a smaller average miss; group sizes are retained.

Source: UCI Adult; unweighted historical sample. Same Week 3 partitions and grouped CV folds.

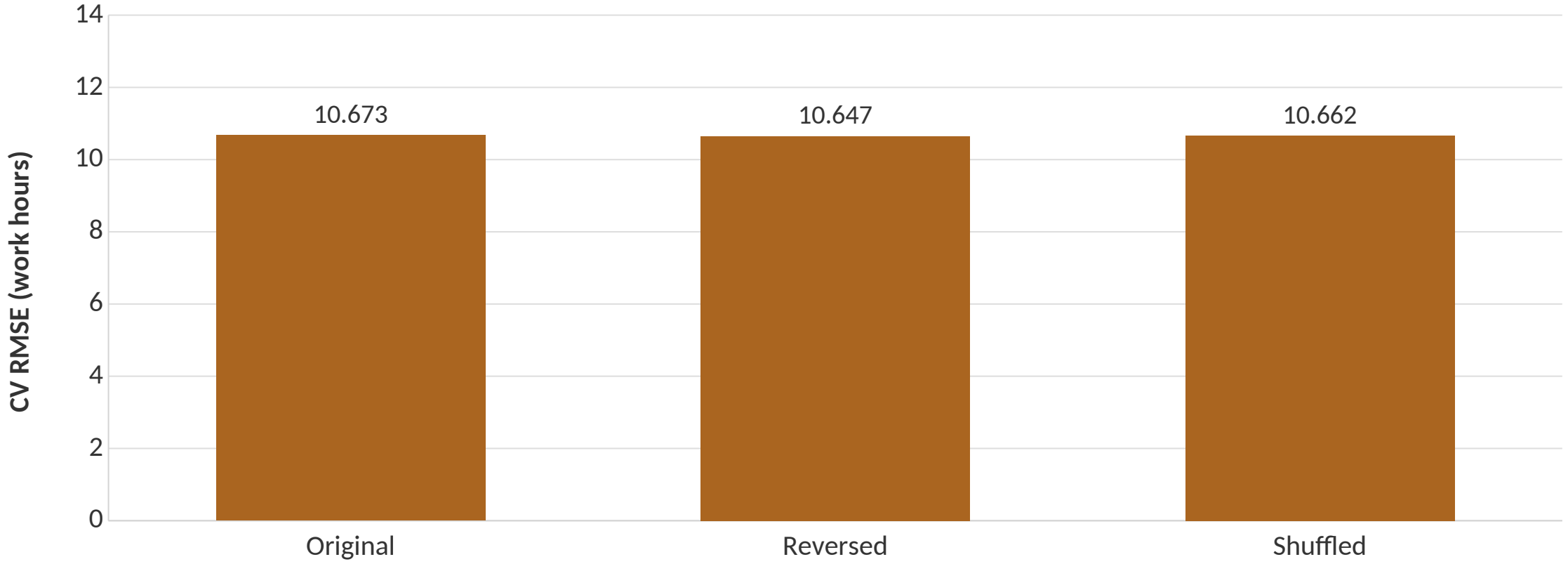
Report: Bias by Workweek Length



Bias = predicted minus actual hours. Positive means overprediction; negative means underprediction. A small overall error can hide large group errors.

Source: UCI Adult; unweighted historical sample. Same Week 3 partitions and grouped CV folds.

Report: RNN Feature-Order Sensitivity



Reversed order scores slightly below MLP's 10.667 CV RMSE. These order diagnostics do not change selection or use test scores. No decisive architecture advantage is established.

Source: UCI Adult; unweighted historical sample. Same Week 3 partitions and grouped CV folds.

Report: Findings and Model Fit

- MLP - expanded was selected by CV. Test RMSE is 10.65 hours and MAE is 7.25 hours.
- With seven inputs, MLP test RMSE (10.65) is lower than RNN (10.78). Expanded SVR has RMSE 10.71.
- MLP - expanded changes RMSE by 2.2% lower versus Week 3 SVR (10.89). All three stated hypothesis criteria are met.
- The RNN has lower short-workweek MAE (11.64 versus 12.62). Expanded SVR has lower overall MAE (7.07 versus 7.25).
- The data, outcome, and validation evidence should guide model choice. Artificial field order and a limited search prevent a general claim that MLP is superior.

Act: Apply the Selected Model

Age	Occupation	Actual hours	Predicted hours
49	Other-service	16	34.58
31	Prof-specialty	50	40.92
23	Protective-serv	52	41.27
30	Adm-clerical	40	43.45
22	Other-service	15	32.42

First five test rows in source order, using MLP - expanded. These examples are not selected for accuracy and do not prescribe individual schedules.

Source: UCI Adult; unweighted historical sample. Same Week 3 partitions and grouped CV folds.

Act: Decision and Next Steps

- Retain MLP - expanded as the preferred method in this comparison. Keep the Week 3 models and mean baseline as reference points.
- Continue investigating errors for short and long workweeks. An average improvement is not enough for individual scheduling or employment decisions.
- Test current independent data, additional seeds and splits, and fairness across relevant groups before considering operational use.
- The same historical test set has already informed this project. Results are internal comparisons; additional information and acceptable-error criteria are still needed.

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Data: https://doi.org/10.24432/C5XW20